

4.12 EXAMPLE 10

MIKROELEKTRONIKA

Using LCD display

This example illustrates the use of an alphanumeric LCD display. The function libraries simplify this program, which means that the effort made to create software pays off in the end.

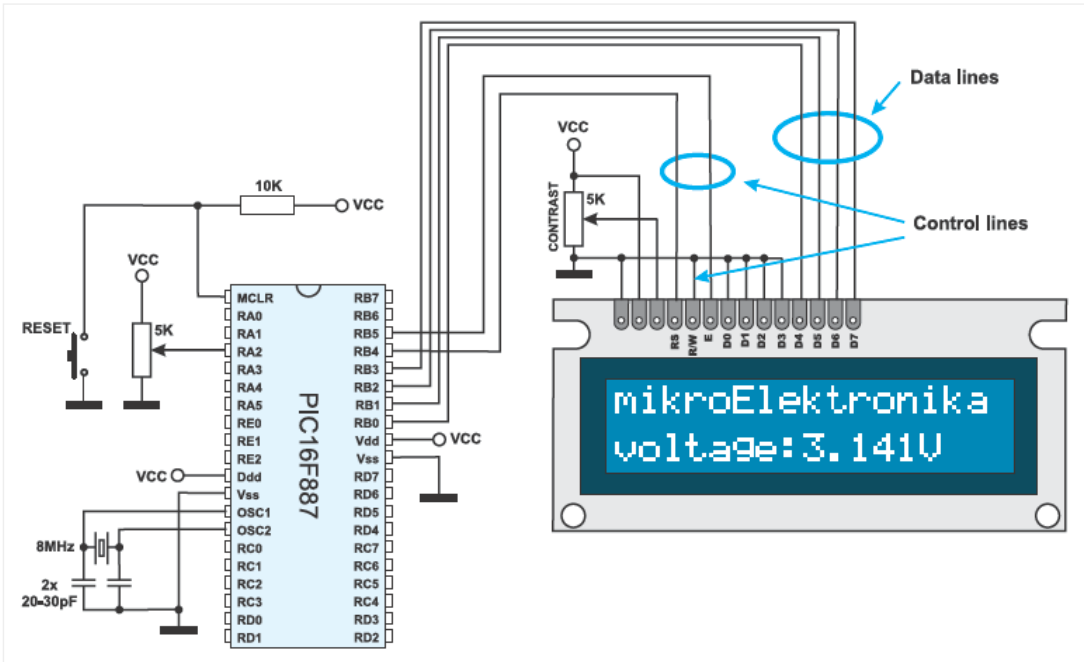
A message written in two lines appears on the display:

mikroElektronika
LCD example

Two seconds later, the message in the second line is changed and displays voltage present on the A/D converter input (the RA2 pin). For example:

mikroElektronika
voltage:3.141V

In true device, the current temperature or some other measured value can be displayed instead of voltage.



In order to make this example work properly, it is necessary to tick off the following libraries in the *Library Manager* prior to compiling:

- ADC
- LCD

```

2
3 // LCD module connections
4 sbit LCD_RS at RB4_bit;
5 sbit LCD_EN at RB5_bit;
6 sbit LCD_D4 at RB0_bit;
7 sbit LCD_D5 at RB1_bit;
8 sbit LCD_D6 at RB2_bit;
9 sbit LCD_D7 at RB3_bit;
10 sbit LCD_RS_Direction at TRISB4_bit;
11 sbit LCD_EN_Direction at TRISB5_bit;
12 sbit LCD_D4_Direction at TRISB0_bit;
13 sbit LCD_D5_Direction at TRISB1_bit;
14 sbit LCD_D6_Direction at TRISB2_bit;
15 sbit LCD_D7_Direction at TRISB3_bit;
16 // End LCD module connections
17
18 unsigned char ch; //
19 unsigned int adc_rd; // Declare variables
20 char *text; //
21 long tlong; //
22
23 void main() {
24     INTCON = 0; // All interrupts disabled
25     ANSEL = 0x04; // Pin RA2 is configured as an analog input
26     TRISA = 0x04;
27     ANSELH = 0; // Rest of pins are configured as digital
28
29     Lcd_Init(); // LCD display initialization
30     Lcd_Cmd(_LCD_CURSOR_OFF); // LCD command (cursor off)
31     Lcd_Cmd(_LCD_CLEAR); // LCD command (clear LCD)
32
33     text = "mikroElektronika"; // Define the first message
34     Lcd_Out(1,1,text); // Write the first message in the first line
35     text = "LCD example"; // Define the second message
36     Lcd_Out(2,1,text); // Define the first message
37
38     ADCON1 = 0x82; // A/D voltage reference is VCC
39     TRISA = 0xFF; // All port A pins are configured as inputs
40     Delay_ms(2000);
41
42     text = "voltage:"; // Define the third message
43
44     while (1) {
45         adc_rd = ADC_Read(2); // A/D conversion. Pin RA2 is an input.
46         Lcd_Out(2,1,text); // Write result in the second line
47         tlong = (long)adc_rd * 5000; // Convert the result in millivolts
48         tlong = tlong / 1023; // 0..1023 -> 0-5000mV
49         ch = tlong / 1000; // Extract volts (thousands of millivolts)
50         // from result
51         Lcd_Chr(2,9,48+ch); // Write result in ASCII format
52         Lcd_Chr_CP('.');
53         ch = (tlong / 100) % 10; // Extract hundreds of millivolts
54         Lcd_Chr_CP(48+ch); // Write result in ASCII format
55         ch = (tlong / 10) % 10; // Extract tens of millivolts
56         Lcd_Chr_CP(48+ch); // Write result in ASCII format
57         ch = tlong % 10; // Extract digits for millivolts
58         Lcd_Chr_CP(48+ch); // Write result in ASCII format
59         Lcd_Chr_CP('V');
60         Delay_ms(1);
61     }
62 }

```



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